

Name: Hossein Golpayegani	Labor Protokoll	Datum: 05.05.2025
Klasse: 6ACIF		Übung Nr.: 5
Lehrer: Prof. Spanring & Horny	Abgabedatum: 05.05.2025	Raum Nr.: Platz Nr.: Bewertung:



Packet Tracer - Configure IPv6 Addressing

Schritt 1: IPv6-Routing aktivieren

```
enable  
configure terminal  
ipv6 unicast-routing
```

Schritt 2: G0/0 konfigurieren

```
interface gigabitEthernet 0/0  
ipv6 address 2001:db8:1:1::1/64  
ipv6 address fe80::1 link-local  
no shutdown  
exit
```

Schritt 3: G0/1 konfigurieren

```
interface gigabitEthernet 0/1  
no ipv6 address 2001:db8:1:1::1/64 ! (falls fälschlich gesetzt)  
ipv6 address 2001:db8:1:2::1/64  
ipv6 address fe80::1 link-local  
no shutdown  
exit
```

Schritt 4: S0/0/0 konfigurieren

```
interface serial 0/0/0  
ipv6 address 2001:db8:1:a001::2/64  
ipv6 address fe80::1 link-local  
no shutdown  
exit
```

```
R1
CLI
IOS Command Line Interface

Press RETURN to get started!

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 uni?
unicast-routing
R1(config)#ipv6 uni?
unicast-routing
R1(config)#ipv6 unicast-routing
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ipv6 address 2001:db8:1:1::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
R1
CLI
IOS Command Line Interface

R1(config-if)#interface gigabitEthernet 0/1
R1(config-if)#no ipv6 address 2001:db8:1:1::1/64
R1(config-if)#ipv6 address 2001:db8:1:2::1/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ipv6 address 2001:db8:1:a001::2/64
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
```

R1

CLI

IOS Command Line Interface

```
R1(Config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ipv6 interface brief
GigabitEthernet0/0          [up/up]
    FE80::1
    2001:DB8:1:1::1
GigabitEthernet0/1          [up/up]
    FE80::1
    2001:DB8:1:2::1
GigabitEthernet0/2          [administratively down/down]
    unassigned
Serial0/0/0                 [up/up]
    FE80::1
    2001:DB8:1:A001::2
Serial0/0/1                 [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
R1#
R1#show running-config
Building configuration...

Current configuration : 996 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
ip cef
ipv6 unicast-routing
!
```

Copy

Paste

R1

CLI

IOS Command Line Interface

```
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:1:1::1/64
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:1:2::1/64
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
no ip address
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:1:A001::2/64
!
interface Serial0/0/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
```

Accounting

Desktop Programming

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address 2001:DB8:1:1::4 / 64

Link Local Address FE80::201:C7FF:FE83:3CED

Default Gateway FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

CAD

Desktop Programming

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address 2001:DB8:1:2::4 / 64

Link Local Address FE80::20B:BEFF:FE8E:73E2

Default Gateway FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Design

Desktop
Programming

IP Configuration
X

Interface
FastEthernet0

IP Configuration

☐ DHCP
☒ Static

IPv4 Address

Subnet Mask

Default Gateway
0.0.0.0

DNS Server
0.0.0.0

IPv6 Configuration

☐ Automatic
☒ Static

IPv6 Address
2001:DB8:1:2::2 / 64

Link Local Address
FE80::230:A3FF:FE02:76B

Default Gateway
FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication
MD5

Username

Password

Engineering

Desktop
Programming

IP Configuration
X

Interface
FastEthernet0

IP Configuration

☐ DHCP
☒ Static

IPv4 Address

Subnet Mask

Default Gateway
0.0.0.0

DNS Server
0.0.0.0

IPv6 Configuration

☐ Automatic
☒ Static

IPv6 Address
2001:DB8:1:2::3 / 64

Link Local Address
FE80::209:7CFF:FE61:6505

Default Gateway
FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication
MD5

Billing

Desktop

Programming

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

Subnet Mask

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

2001:DB8:1:1::3

/ 64

Link Local Address

FE80::207:ECFF:FE29:D35A

Default Gateway

FE80::1

DNS Server

802.1X

Sales

Desktop

Programming

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

Subnet Mask

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

2001:DB8:1:1::2

/ 64

Link Local Address

FE80::201:97FF:FE13:ECE0

Default Gateway

FE80::1

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

LAB4_2_(Teil 2).doc

Seite 6

Design

Desktop Programming

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:1:a001::1

Pinging 2001:db8:1:a001::1 with 32 bytes of data:

Reply from 2001:DB8:1:A001::1: bytes=32 time=34ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=17ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254

Ping statistics for 2001:DB8:1:A001::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 34ms, Average = 13ms

C:\>
```

Sales

Desktop Programming

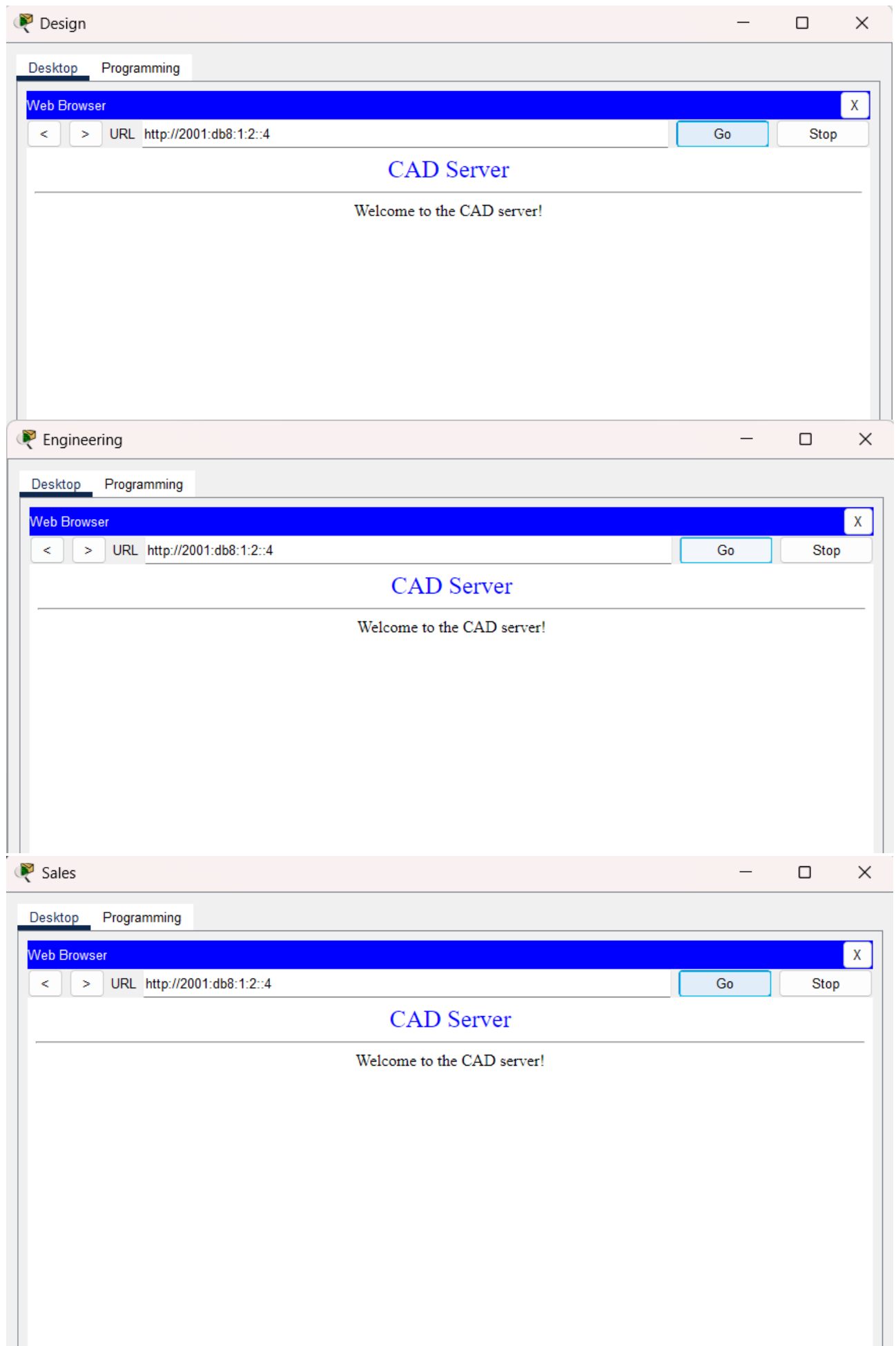
Command Prompt

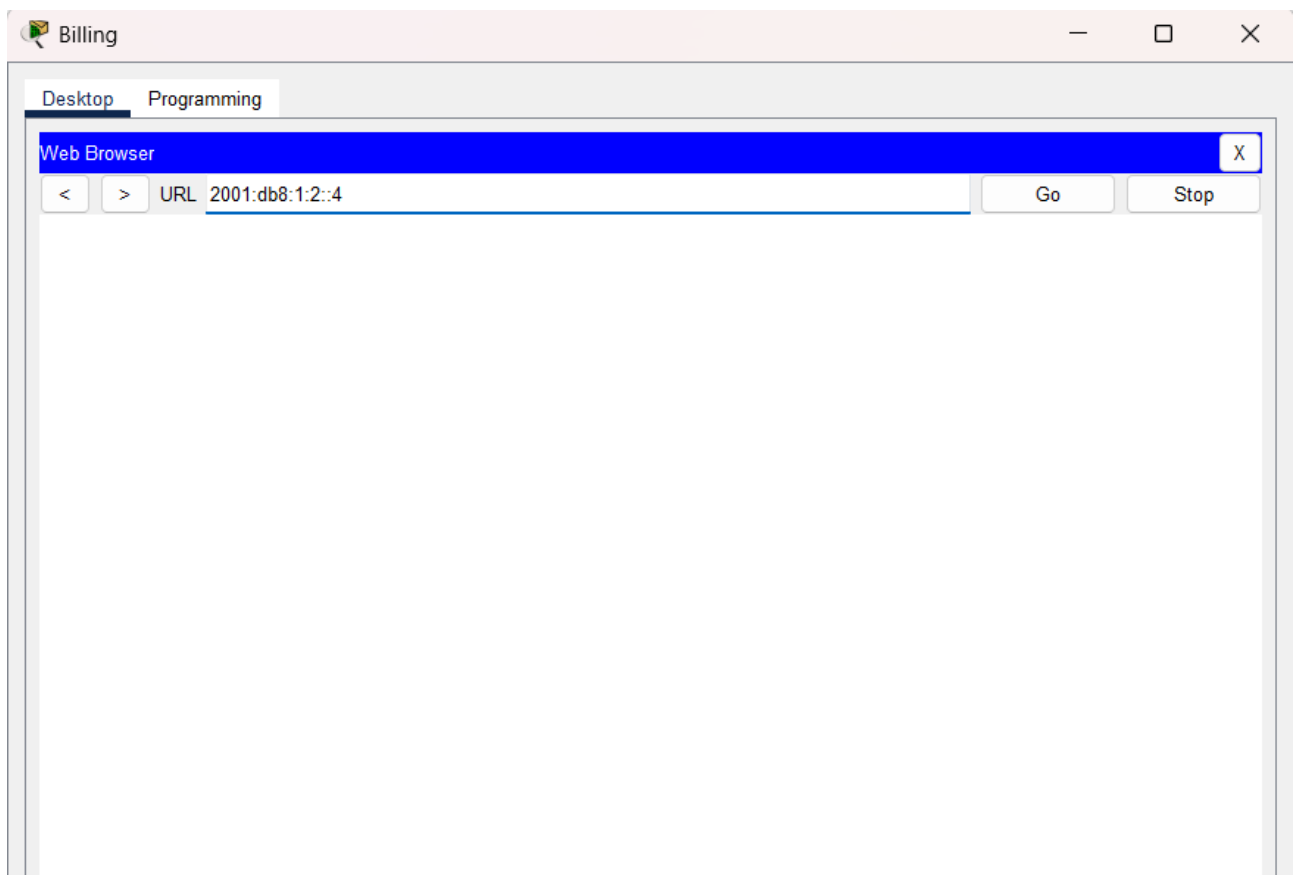
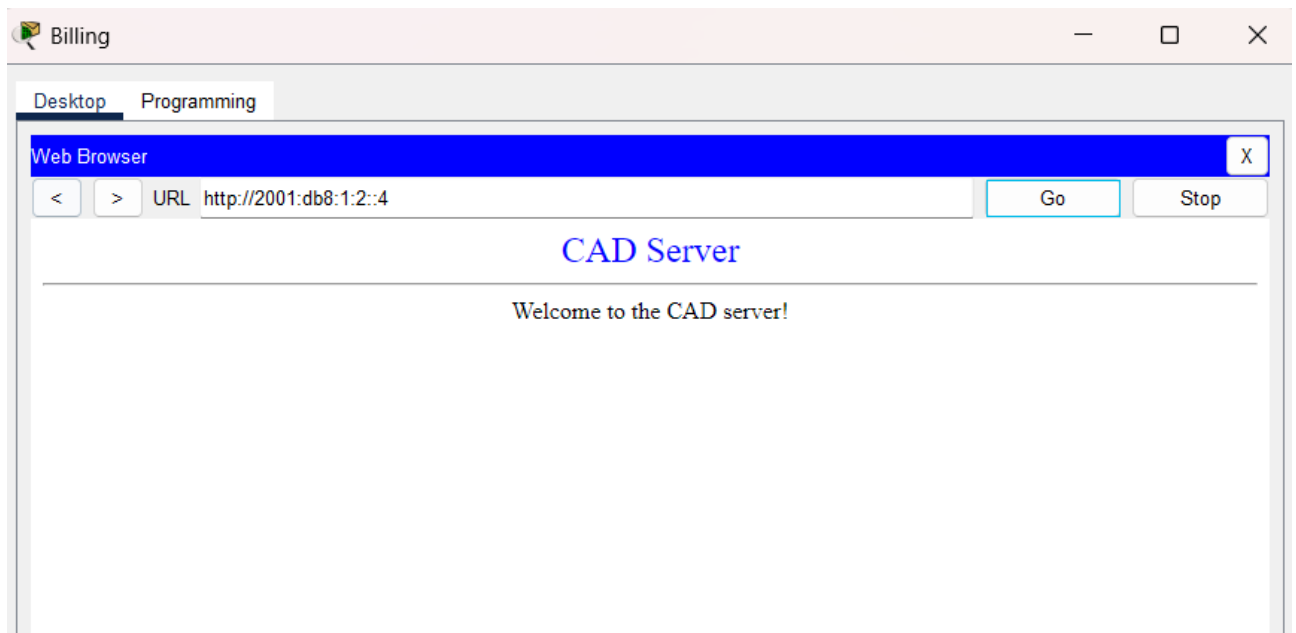
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:1:a001::1

Pinging 2001:db8:1:a001::1 with 32 bytes of data:

Reply from 2001:DB8:1:A001::1: bytes=32 time=20ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=1ms TTL=254
Reply from 2001:DB8:1:A001::1: bytes=32 time=28ms TTL=254

Ping statistics for 2001:DB8:1:A001::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 28ms, Average = 12ms
```





Cisco Packet Tracer - C:\Users\hgolp\Desktop\12.6.6 Packet Tracer - Configure IPv6 Addressing.pka - Hossein Golpayegani - 2025-05-05 17:15:15

File Edit Options View Tools Extensions Window Help

Activity Results

Congratulations Hossein Golpayegani! You completed the activity.

Overall Feedback Assessment Items Connectivity Tests

Congratulations! You successfully completed the **Packet Tracer - Configuring IPv6 Addressing** activity.

